

Yugal Joshi

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EDUCATION

M.S. Research in Data Science & Artificial Intelligence - IIT Madras July 2026 – present

B.E. (Hons.) Computer Science-BITS Pilani,Pilani Aug 2020 – June 2024

Relevant Courses: *Data Structures & Algorithms,Databases, OOP, Operating System, Computer Networks, Computer Graphics*

Govt. Inter College Peora, Nainital Apr 2017 – June 2020

Senior Secondary (12th), Uttarakhand State Board **Grade : 95.2/100**

Secondary (10th), Uttarakhand State Board **Grade : 97.0/100**

TECHNICAL SKILLS

Programming Languages: *Java, Python, C++, C, SQL.*

Frameworks/Libraries/DBs/Tools: *Spring Boot, PySpark, Redis (streams + DB),Maria DB, Kafka, PostgreSQL, Cassandra (NoSQL), Spring Batch, GitLab, GitHub, Postman, Shell Scripting*

WORK EXPERIENCE

Software Engineer - Mindgate Solutions, Mumbai Aug-2024 - Aug 2025

- Scaled Two-way *reconciliation system* from **1M to 100M transactions/day**, optimizing memory by **60%** and boosting performance by **50%** via *distributed computing, dynamic load balancing, and on-demand partitioning*.
- Revamped Mindgate's payout pipeline, reducing processing time by **50%** and increasing transaction volume **5x** via **Redis Streams & Cassandra**.
- Built a Bash script enabling versioned JAR deployment with rollback, dependency checks, and zero-downtime releases across environments.
- **Technologies used:** *Spring Boot, PySpark, Redis, PostgreSQL, Cassandra, Python, Shell Scripting.*

RESEARCH/TECHNICAL PROJECTS

Working on Correlated Noise Mechanism for Learning Problems (Differential Privacy)

Supervised by Dr. Krishna Pillutla, IIT Madras

ERPLAG Compiler (C/C++)

Supervised by Dr. Shashank Gupta, BITS Pilani

- Developed a compiler in C which compiles x86 NASM assembly for a Turing-complete language-ERPLAG.
- Implemented unique features like dynamic arrays, recursion, and functions supporting multiple return values.

Hunter Ops (Unity)

Supervised by Dr. Sundaresan Raman, BITS Pilani

- Developed a third-person shooter game in C# with missions, day-night cycles, and inventory for immersive gameplay.
- Utilized Blender for crafting high-quality game assets and integrated them with Unity for fluid animations.

ACHIEVEMENTS

- Awarded **Avery Dennison Innovation Award** worth **Rs 100000** for being in the **top 10 nationwide** in Avery Dennison's innovation challenge organised by Institute of International Education (IIE) in 2022
- Achieved the **1st Rank** in Uttarakhand across all academic streams out of 200000+ students in Junior College (12th).